

RVRC Symposium 2027

Beyond Borders: Shaping Sustainable Futures Together

Poster Gallery

NUS National University of Singapore | RVRC Ridge View Residential College National University of Singapore



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Beyond Borders: Shaping Sustainable Futures Together marks a new chapter for the annual RVRC Symposium, expanding into a regional platform for knowledge exchange and collaboration. Bringing together students, educators, and industry partners from Singapore and across Asia, the Symposium explores sustainability and workplace readiness through research, shared experiences, and interdisciplinary dialogue. Organised around two complementary threads, the Symposium encourages participants to exchange perspectives, address real-world challenges, and foster meaningful collaborations towards building a more inclusive, resilient, and sustainable future.

Thread One

ReSynergy: Reimagining resilient futures

Focused on environmental sustainability in Singapore and across the region, presentations in this thread explore the nexus between biodiversity conservation and sustainable development, while examining how policymaking can strengthen community and environmental resilience. The thread further showcases student workplace readiness through internships or industry-partnered projects that provide authentic learning experiences beyond the classroom. By applying their knowledge to real-world challenges, students are empowered to think critically, develop professional skills, and develop their capacity to drive meaningful change in support of sustainable development goals.

Thread Two

InSynergy: Co-creating inclusive futures

Highlighting how heritage and culture shape communities in a rapidly changing world, this thread explores the importance of developing soft skills and cultural awareness in building a more equitable shared future. Through conversations on cultural expressions and preservation, this thread aims to foster meaningful interactions between people and places while deepening understanding of the diverse perspectives across ASEAN. The thread also showcases student workplace readiness through internships or projects with industry and community partners, where students develop professional, communication, and intercultural competencies while applying their learning to real-world contexts.



Interactive Digital Poster Abstracts

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Navigating Complexity: Lessons in Sustainability Practice from a Healthcare Internship

Yasmin Binte Noh

This poster presents findings from my internship in the healthcare industry with the SingHealth sustainability team, examining how real-world sustainability work develops professional competencies that extend far beyond classroom learning. Key insights are organised across thematic panels spanning stakeholder coordination, data governance, and behavioural change, to invite dialogue on the tensions between technical and relational dimensions of sustainability practice. As an intern supporting Scope 1 and 2 greenhouse gas emissions reporting, I encountered the operational realities of sustainability management, particularly the challenge of consolidating fragmented data across multiple public health institutions with disparate equipment, data ownership structures, and reporting practices. This experience revealed that sustainability work is fundamentally a coordination exercise, requiring stakeholder engagement, organisational navigation, and professional communication rather than technical expertise alone. Drawing on the concept of Corporate Change Agents for Sustainability (Schaltegger et al., 2023, 2024), this poster illustrates how sustainability professionals function as facilitators and coordinators of systemic change. Direct engagement opportunities with local stakeholders during my internship further reinforced the importance of systems thinking and behavioural competencies in addressing interconnected environmental challenges. As I wrap up a co-authored paper from the findings of this internship, these experiences underscore the transferable value of authentic workplace learning for undergraduates in cultivating the adaptive, cross-disciplinary capabilities needed to advance sustainable development goals.

Seeing Reefs in 3D: Accessible Photogrammetry for Post-Bleaching Coral Reef Monitoring in Kota Kinabalu, Sabah, Malaysia

Timothy Ong Zhan Fei

The coral reefs of Kota Kinabalu face mounting pressures from coastal development, sedimentation, and climate-induced thermal stress, underscored by the 2024 global bleaching event. While effective monitoring is critical for safeguarding marine biodiversity, traditional surveys often miss fine-scale structural details, and advanced scanning technologies remain costly. This project explores the potential of underwater 3D photogrammetry using consumer-grade action cameras as an accessible, non-invasive method for assessing reef condition and recovery. By capturing overlapping imagery across 10 survey sites in Kota Kinabalu, the workflow reconstructs high-resolution 3D models and orthomosaics. These digital assets allow researchers to accurately quantify live coral cover, analyze community composition, and examine post-bleaching taxonomic responses without disturbing fragile habitats. Integrated with machine-learning-assisted spatial annotation, the process streamlines analysis to detect patterns of bleaching, physical degradation, and early recruitment, creating an enduring digital baseline. By transforming accessible equipment into reliable analytical tools, this approach provides a practical framework for local monitoring teams, supporting proactive reef management and strengthening regional environmental resilience.

Sustainable Healthcare: Elderly Medication Safety and Career Readiness

Chen Ian In

Population ageing is reshaping healthcare delivery, long-term care, and sustainable urban wellbeing across East Asian societies, especially in Macao. In Macao, many community-dwelling older adults face multimorbidity and polypharmacy, struggling to interpret medication information, distinguish similar packaging, maintain complex regimens, and communicate health histories during clinical or telehealth consultations. This project introduces Medication Safety Companion, an ageing-friendly AI-enabled platform fostering inclusive, sustainable healthy ageing through responsible innovation. Integrating wearable first-person visual sensing, intelligent text recognition, voice-based interaction, retrieval-augmented knowledge support, and privacy-preserving data management, it assists older adults with medication verification, personalized reminders, pharmaceutical information access, and structured healthcare communication. A human-audited pharmaceutical knowledge framework is incorporated to enhance reliability and minimize risks associated with AI-generated misinformation. By curbing medication mismanagement, the platform also drives sustainable healthcare. Beyond individual safety, this initiative strengthens community resilience by lowering technology and accessibility barriers, supporting intergenerational caregiving, and promoting sustainable practices. The project also embodies student workplace readiness: undergraduates work alongside healthcare professionals, developers, and community organizations, applying classroom knowledge to real-world challenges, cultivating critical thinking and professional skills while contributing to sustainable development and informing policy for age-friendly, environmentally responsible communities.

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From Classroom Learning to Community Empowerment: Advancing Disability Inclusion through Social Enterprise

Kuah Sze Min, Shermaine

Inspired by the course's focus on disability awareness, accessibility, advocacy, and inclusive participation, Shermaine pursued an internship with a social enterprise providing training for persons with disabilities. Using an interactive screen, the poster will feature photos and visual highlights from Shermaine's course project and internship experience, illustrating how classroom learning was extended into real-world community engagement. Through the internship, Shermaine contributed to efforts that support persons with disabilities in developing skills, confidence, and meaningful pathways for participation. Her journey reflects the spirit of InSynergy, where inclusive futures are co-created through partnerships between individuals, communities, and organisations. Rather than viewing inclusion as a concept confined to coursework, Shermaine translated learning into purposeful action. This experience demonstrates workplace readiness, social responsibility, and a growing identity as an advocate for the disability community.

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How does food culture in Borneo vary across time and space?

Regine Koh Wei Shan, Phoon Yu Ying, Poh How Ee, Dylan, Tan Teong Kai, Nathan Tew Ci Heng

This interactive poster is based on the students' engagements and experiences during the study trip to East Malaysia. The poster presents their insights on the topic of food culture in Borneo by looking at how it is shaped by the pressures of conservation, commercialisation and urbanisation. It also examines the balance between adaptation and preservation through the diverse perspectives and examples from different communities in Sabah and Sarawak.

Improving Women's Access to Small Indigenous Fish through a Gender-Responsive Mola Gill Net System in Rural Bangladesh.

Wismen A Bendula

Aquaculture and fisheries play important roles in Bangladesh's socio-economic development, producing 5.02 million tonnes of fish and contributing approximately 2.5% to national Gross Domestic Product (GDP) in 2024. Most aquatic foods are consumed domestically, with per capita daily fish intake reaching 67.8g in 2022. Bangladesh has made substantial progress in reducing hunger and malnutrition, and is self-sufficient in rice, fish, and vegetables. However, micronutrient deficiencies persist, particularly among vulnerable groups such as women and children. Nutrition-sensitive pond polyculture systems, in which small indigenous fish species (SIS), such as mola carplet (Amblypharyngodon mola), are cultured alongside carps, have been introduced to enhance household food and nutrition security. Although evidence demonstrates the nutritional benefits of pond polyculture, barriers remain to women's access to ponds and their participation in harvesting the fish for household consumption. To address this constraint, a low-cost, gender-specific mola gill net system was introduced and evaluated for its efficiency in facilitating fish harvesting and improving women's access to SIS. This study assesses the potential of the technology to strengthen women's participation in small-scale aquaculture and opportunities to scale in other countries.



Poster Abstracts

Beyond Targets: Scenario Modelling Universiti Malaya's Pathway to Carbon Neutrality and Net Zero

Muhammad Daniel bin Md Lazim

Universities across ASEAN are making public climate commitments, yet few can demonstrate whether their planned initiatives will actually deliver them. Universiti Malaya (UM) has pledged carbon neutrality by 2030 and net zero by 2050, but its institutional planning documents set targets without quantified reduction trajectories or scenario testing. This study addresses that gap by developing a scenario model of UM's operational greenhouse gas (GHG) emissions using the Low Emissions Analysis Platform (LEAP). Following the GHG Protocol, the model covers Scope 1 (stationary combustion, mobile combustion and fugitive emissions) and Scope 2 (purchased electricity). It establishes a baseline inventory, identifies emissions hotspots, and projects alternative decarbonisation pathways to 2050, ranging from business-as-usual to accelerated measures such as energy efficiency, on-campus solar and national grid decarbonisation. These pathways are then tested against UM's 2030 and 2050 targets to evaluate whether existing initiatives are sufficient and what additional action would be required. The work builds on the author's emissions data compilation and GHG calculations for UM's Carbon Neutrality Acceleration project. The study offers a transferable approach for regional universities to turn climate pledges into accountable, measurable pathways.

Effect of underwater soundscape to colonization of hard bottom community in Teluk Aling, Penang National Park.

Najwa Hannani Binti Mohd Yusoff

Anthropogenic marine activities, including piling, coastal reclamation, dredging, and vessel traffic, generate underwater noise that can disrupt sensory cues, alter organismal behaviour, and potentially influence larval recruitment and community development. This study investigates how underwater soundscapes influence the colonisation of hard-bottom communities at Teluk Aling, Penang. Two experiments were conducted using PVC settlement panels (15 × 15 cm). In the first experiment, 16 panels were exposed to vessel-traffic sounds and 16 to natural ambient sound underwater. In the second, a similar experimental design compared panels exposed to a healthy reef soundscape with those exposed to natural ambient sound. PVC panels were retrieved after four weeks for wet-weight measurement and photographic documentation, with species composition, abundance, and percentage organism cover subsequently quantified. Panels were then redeployed for a further four weeks before the next assessment. Preliminary results indicate that panels exposed to the healthy-reef soundscape exhibited greater organism colonisation and higher species richness than ambient controls, whereas panels exposed to anthropogenic vessel noise showed lower species richness. These findings suggest that soundscape enrichment may offer a promising nature-based approach to support ecological restoration, while reducing anthropogenic noise may enhance the resilience of coastal ecosystems.

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How could traditional fishing culture survive differing stakeholder interests in Kelab Alami?

Chye Pin Ke, Lim Yi Hui, Pang Wei Quan Wilber, See Wei Shan, Wee Wye Keong

In Malaysia, fishing culture sustains the social bonds, traditional ecological knowledge, and intergenerational livelihoods that hold coastal communities together despite economic and environmental pressures (Obi et al., 2025). In Mukim Tanjung Kupang, this is reflected through traditional fishing practices shaped by generations of lived experiences, including intimate understanding of currents, fish behaviour, and seasonal cycles. Beyond knowledge, it also sustains the community identity, which is a representation of Malaysia's rich cultural heritage. As this fishing culture faces growing pressure, multiple stakeholders have become involved in shaping its future, often with competing priorities and visions of sustainability. This analysis therefore examines how these differing interests can be reconciled in ways that preserve fishing culture on the community's own terms.

Life on the Surface: Cataloguing the semi-aquatic true bugs (Hemiptera: Gerromorpha) for ecosystems monitoring and bioresource management in the Philippine Archipelago

Shaira Aiko A. Valmeo

Despite being a biodiversity hotspot with high levels of endemism, the Philippine biodiversity is at risk from overlapping threats that may originate from natural causes or from various anthropogenic factors. As a result, taxonomists find themselves "racing against time" as species may disappear before they are formally identified, but this is further burdened by a severe shortage of expertise and resources in the Global South. Hence, this study provides baseline data in the study of freshwater ecosystems and marine-associated habitats, promoting knowledge about life below water and on land. Infraorder Gerromorpha Popov, 1971 (Insecta: Hemiptera: Heteroptera) is a taxonomic group composed of semi-aquatic true bugs, including water striders and riffle bugs. Despite harboring high endemism of true bugs, the diversity and distribution of Gerromorpha in the Philippines remain understudied. Thus, this study seeks to update the checklist of gerromorphs across the Philippine biogeographic regions to develop a policy framework for freshwater conservation. Overall, this study primarily targets UN SDGs 14 Life Below Water and 15 Life on Land, while also contributing to other SDGs focusing on education (SDG 4) and partnerships (SDGs 16, 17). Ultimately, the goal of the project is to translate the taxonomic data into actionable outcomes by providing proper scientific knowledge as a basis for conservation policies at the local level, delivering clear assessments of biodiversity across taxa, and empowering local communities through targeted training and capacity-building workshops.

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Rhetoric in the Evolution of Singapore's Electric Vehicle Policy

Lee Yu Yang

This paper examines the Singapore government's decade-long hesitation and subsequent 2020 policy pivot regarding electric vehicle (EV) adoption. Moving beyond traditional technical and economic explanations, it argues that the state's initial reluctance was a deliberate social construction of risk. Using an integrated framework of Critical Discourse Analysis (CDA) and framing theory, the study analyzes official government communications and state-aligned news articles to deconstruct the rhetoric surrounding EV policy. Findings show that between 2011 and 2019, the government framed EV adoption as an existential threat to national stability, casting charging infrastructure deployment as an unsolvable social conflict, EV ownership as an ideological risk to car-lite public transit goals, and electrification as an incomplete environmental solution. Following the 2020 Singapore Green Plan announcement, these same risks were swiftly reconstructed into manageable national targets. This shift demonstrates how risk framing serves as a tool of technocratic statecraft to justify policy inaction, delay free-market disruptions, and maintain centralized control until top-down execution is viable. The paper concludes that this strategy provides a predictive model for how Singapore will navigate future disruptive technologies.

SL2i

Lu Yiying, Cheerly Tannia Hartono

SL2 Impact (SL2i) is a ground-up community that promotes environmental stewardship and community resilience through repair and reuse initiatives. Despite its positive environmental and social contributions, SL2i faces challenges in sustaining participation and generating stable revenue. This poster shares results from a survey conducted to understand public interest in handyman training programmes and identify opportunities for improvement. Results indicate stronger demand for shorter, affordable, and practical courses. Based on these findings, we recommend redesigning course offerings, enhancing promotional efforts, and developing new engagement strategies to attract participants. These measures can improve SL2i's financial sustainability while enabling the organisation to continue delivering meaningful environmental and social benefits to the community.

Artscapes, Eco-Art, and Sense of Place

Hathwar Abhay, Yu Jiaxin, Lee Wan Xin, Lu Yiying

Rapid urbanisation increasingly severs the bond between residents and the spaces they inhabit, eroding the sense of place that underpins environmental stewardship and community resilience. This project investigates how placemaking interventions grounded in artscape and ecological art principles can restore that connection within car-centric urban environments, in direct support of SDG 11 (Sustainable Cities and Communities) and SDG 15 (Life on Land). Drawing on established literature linking sense of place to stewardship behaviour, we worked on a site that is visually incoherent, ecologically sparse, and socially underutilised. As part of an actual upcoming urban intervention, we developed a suite of site-specific, monumental-scale design proposals to transform a passive slope into an active visual landmark legible to both motorists and pedestrians. In our proposal, integrated vegetation enhances local biodiversity and softens hard infrastructure, while heritage and cultural references strengthen institutional identity, together advancing SDG 4 (Quality Education) by embedding environmental literacy within the built environment itself. These live interventions aim to cultivate meaningful lived experiences, invite dwelling, and foster emotional attachment to place. Upon implementation, we anticipate measurable improvements in threshold legibility, greater pedestrian dwell time, and enriched ecological value along the corridor. Beyond the site, working on a real-world commission has fundamentally shaped how we approach problem solving. We have learnt how to navigate the complexity of balancing aesthetic ambition with ecological responsibility and genuine community need. We hope our work models how youth, empowered through authentic project experiences, can become active contributors to the broader sustainable urban future in Asian cities.

Finding Common Ground

Gao Yilun, Ting Pin Hao, Chow Jeng Han, Sun Zhuoran, Kaung Khant Zaw

Our presentation is on the concept of privilege in relation to race in Singapore. Specifically, we will be focusing within the field of education to provide deeper insight into how racial privilege can be seen in daily life, and because the audience of students will likely be able to relate personal experiences to the topic. We will be looking at this through the concept of intersectionality, e.g. how race and other social identities such as class and religion intersect, and how that creates privileges for certain groups.

Food as Intersection

Darren Ong Yan En, Chow Jeng Weng, Seth Lee Ying Hong, Nerissa Chou Zi Yuan

Singapore's multicultural heritage — shaped by Chinese, Malay, Indian, and Eurasian communities — finds its most vivid expression in food. This presentation explores how the meeting of diverse cultures has produced fusion cuisine as a natural product of everyday cultural exchange. We examine how fusion food has rejuvenated Singapore's hawker centres, inspiring a new generation of hawkerpreneurs to reimagine traditional dishes and keep the hawker trade alive. At the same time, we ask a critical question: does fusion food preserve local hawker culture, or does it undermine the heritage it draws from?

From Mobility to Meaningful Action: Youth Leadership for Resilient and Sustainable Futures

Mashitoh Binti Abu Bakar

Building resilient and sustainable futures requires young people who can lead, collaborate, adapt, and respond constructively to change. This presentation examines how student leadership and international mobility can develop these capacities within the context of Borneo and the wider ASEAN region. It adopts an autoethnographic approach, using personal experience as a student leader and leader of a 2024 international student mobility delegation to Hatyai, Thailand, to connect individual learning with broader questions of youth leadership, community resilience, and regional engagement. The analysis is further informed by Schön's concept of reflective practice, particularly reflection-in-action. The discussion is organised around three dimensions of learning: cognitive shifts, behavioural action, and interpersonal dynamics. These experiences suggest that mobility can move students beyond passive participation by encouraging greater confidence in contributing ideas, engaging peers, and responding to unfamiliar situations. The presentation concludes that short-term mobility is most valuable when treated as a structured learning process rather than a stand-alone travel experience. Universities can strengthen its impact through pre-departure leadership preparation, interdisciplinary participation, guided reflective practice, and post-mobility community or peer-led initiatives that translate experience into sustained action. Such design can help aspiring youth leaders convert temporary intercultural exposure into sustained campus engagement, volunteerism, and regional collaborative problem-solving.

How Community-based Tourism Impacts the Cultural Identity of Local Communities

Zachary Pek, Joel Rhys Chee, Tee Ren Wey, Charmaine Kwek, Nerissa Chou Zi Yuan

Tourism contributes 15.1% to Malaysia's GDP and supports 3.5 million jobs (Department of Statistics Malaysia, 2025), making it central to national development. Malaysia's population comprises numerous ethnic communities like the Bumiputera, Chinese and Indian communities (Department of Statistics Malaysia, 2020), each with their own cultural identity. In our project, cultural identity is understood as "identification with, or sense of belonging to, a particular group" constructed through shared collective knowledge such as traditions, heritage, language, aesthetics, norms and customs (Chen, 2014). These identities are especially varied and significant in Malaysia's multicultural society. However, communities such as those served by Kelab Alami face growing pressure as coastal development and habitat destruction threaten traditional fishing livelihoods, gradually eroding the shared practices that hold cultural identity together (Rahman, 2020). In response to these pressures, community-based tourism (CBT) has emerged as an approach that places local culture at the centre of tourism activities by involving visitors directly in community life through experiences such as village-style living and participation in daily practices (Tourism Malaysia, 2026.; Kayat, 2010). While CBT may help preserve traditions and strengthen community pride, it can also turn culture into a tourism product shaped to pander to visitor expectations. Hence, this project investigates how community-based tourism affects the cultural identity of local communities.

How Does Urbanisation Affect the Preservation of Cultural Heritage? From Cultural Loss to Adaptive Preservation

Charmaine Chieng Jia Yee, Teng Tzen Xiang Justin, Loo Wei En, Wang Zi Lun, Zachary Teo

This poster examines how urbanisation affects the preservation of cultural heritage, challenging the common assumption that preservation requires traditions to remain unchanged. Drawing on the concept of the cultural iceberg, we argue that cultural heritage can continue to be preserved even when its visible forms evolve, provided that its deeper meanings, values, identities, and social relevance remain intact. Using three Malaysian case studies, Traditional Chinese Medicine (TCM), Zapin dance in Johor, and Think City's urban regeneration initiatives, we explore how urbanisation simultaneously creates pressures and opportunities for cultural continuity. Our findings show that urbanisation may alter the material expression of culture through regulation, modernisation, changing demographics, commercialisation, and redevelopment. However, these changes do not necessarily equate to cultural loss. While tensions surrounding authenticity, commodification, and displacement remain, our analysis suggests that adaptation is often essential for cultural survival in modern cities. We conclude that cultural heritage is preserved when its core values endure, even as its observable forms evolve over time.

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How does Food Culture in Borneo Vary Across Time and Space?

Regine Koh Wei Shan, Phoon Yu Ying, Poh How Ee, Dylan, Tan Teong Kai, Nathan Tew Ci Heng

This project explores the evolving food cultures of Borneo, examining how conservation, commercialisation, and urbanisation have shaped culinary traditions across time and space. Drawing on diverse perspectives and examples from communities in Sabah and Sarawak, it highlights the ongoing tensions between cultural preservation and adaptation, offering insights into how traditional food practices evolve in response to changing social, economic, and environmental conditions.